

Supplementary Material

Agentic Personas for Adaptive Scientific Explanations with Knowledge Graphs

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1 Expert Evaluation

1.1 Demographics

All participant demographics, expertise, and domain knowledge are presented in Table 1.

Table 1: Participant demographics, expertise, and domain knowledge.

	Study (n = 11)
Age Range	25–34: 9; 35–49: 2
Primary Background	Life and Health Sciences: 4 CS/AI-heavy w/ Biomedical Experience: 5 Hybrid Computational-Biology: 2
Domain Knowledge	Knowledge Graphs: 4 Experts, 2 Competent, 3 Novice, 2 No Knowledge AI Systems: 1 Expert, 7 Competent, 1 Novice, 2 No Knowledge Molecular Biology: 2 Experts, 8 Competent, 1 No Knowledge

1.2 Qualitative Analysis

We conducted an initial analysis where results reveal substantial heterogeneity in expert preferences that transcends traditional professional categorizations. Two themes consistently emerged that directly align with our evaluation criteria of relevance and completeness, and therefore with core explanatory virtues in scientific explanation:

- **Background shapes relevance:** Participants judged an explanation as relevant when it aligned with the types of evidence they viewed as meaningful. Life-science experts prioritized biomedical plausibility, CS/AI experts focused on logical structure and predicate choice, and hybrid experts

sought a balance between rigor and brevity. These differences reveal that relevance is not uniform but depends on an expert’s disciplinary lens.

- **Completeness is individually calibrated:** Even within the same background, some experts preferred dense mechanistic detail while others favored concise, high-signal paths. This variability shows that completeness is subjective and cannot be captured by static depth settings or role labels. Instead, experts differ in how much detail they consider necessary for an explanation to feel complete.

2 Persona Narratives and Summaries

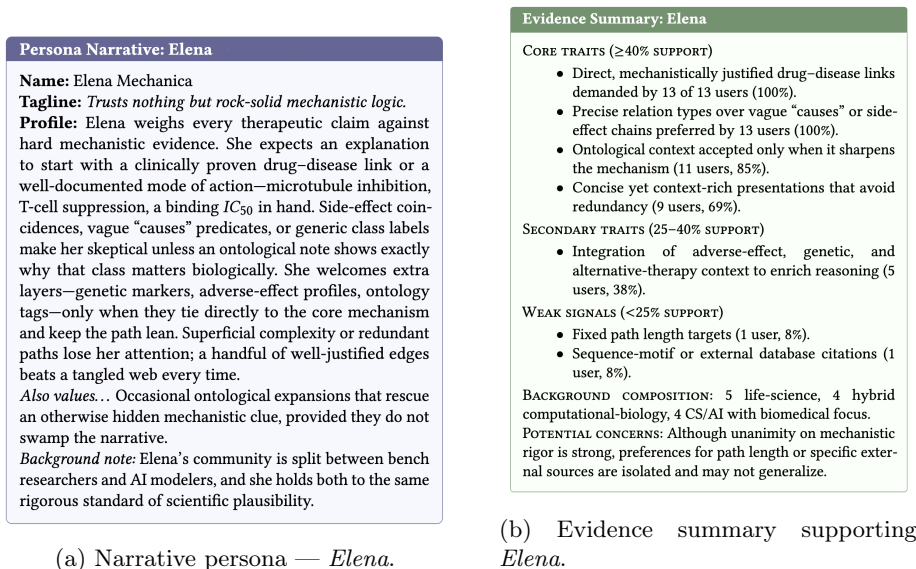


Figure 1: Persona *Elena*: (a) narrative profile and (b) supporting evidence summary.

Persona Narrative: Leo

Name: Leo Insight

Tagline: *Looks for one illuminating link that makes the mechanism click.*

Profile: Leo scans an explanation until a single, definitive connection locks the story into place—an ontological expansion that clarifies a hidden relation or a standout feature like “dependence on opiates” that tags the drug’s true family. Once that anchor appears, he is satisfied so long as the rest of the path stays short and avoids redundant steps. He is comfortable with a couple of alternative paths for completeness, provided each earns its keep and the display never bloats into an unreadable maze.

Also values... Using ontology sparingly to light up indirect relations that a simple “treats” or “causes” link cannot explain.

Background note: Half of Leo’s colleagues come from wet-lab pharmacology, the other half from AI drug-discovery teams, so he stands at the crossroads of formal ontology and data-driven insight.

Evidence Summary: Leo

CORE TRAITS (≥40% SUPPORT)

- Preference for uniquely informative, mechanism-clarifying links (2/2, 100%)

SECONDARY TRAITS (25–40%)

- Judicious ontological expansion for indirect relations (1/2, 50%)
- Limiting path length to avoid complexity (1/2, 50%)

WEAK SIGNALS (<25%)

- None (all mentions ≥50%, small sample size)

BACKGROUND COMPOSITION: 1 life-science, 1 CS/AI with biomedical expertise.

CONCERNS: With only 2 contributors, even shared themes rest on minimal evidence.

(a) Narrative persona — *Leo*.

(b) Evidence summary supporting *Leo*.

Figure 2: Persona *Leo*: (a) narrative profile and (b) supporting evidence summary.

3 Details on Adaptive Training via Curriculum Learning

Full persona evaluation for every path is unnecessary and costly. We therefore adopt a self-paced curriculum that gradually raises the requirements for invoking persona scoring. At training step t , we maintain a relevance reward threshold $\tau_{\text{relevance}}(t)$ that increases over time:

$$\tau_{\text{relevance}}(t) = \begin{cases} 0.50 & \text{if } t < 60, \\ 0.55 & \text{if } 60 \leq t < 80, \\ 0.60 & \text{if } 80 \leq t < 100, \\ 0.65 & \text{otherwise.} \end{cases}$$

The maximum number of training steps is user-defined, with early stopping triggered if performance does not improve for three consecutive evaluations. In our experiments, we used a maximum of 200 steps.

At every step, fidelity is enforced as a prerequisite: only paths that successfully connect the source and target entities are considered for persona reward. Among these faithful paths, we partition further into three tiers based on relevance:

1. **High-relevance** (relevance $> \tau_{\text{relevance}}(t)$): evaluated by the agentic persona; reward is the agentic score $S_{\text{agentic}}^{(i)} \in [0, 1]$.

2. **Medium-relevance** ($0.5 \leq \text{relevance} \leq \tau_{\text{relevance}}(t)$): assigned a modest fixed reward (0.25) to sustain learning.
3. **Low-relevance** ($\text{relevance} < 0.5$): assigned a small exploratory reward (0.10) to avoid premature collapse.

This curriculum lets the agent first master path-finding under lenient criteria, then progressively targets higher-quality paths for fine-grained persona alignment. Beyond cost savings, this design is computationally efficient: rather than asking the LLM to evaluate hundreds of paths per batch where only some might be sufficiently informative, we pre-filter using relevance scores to identify the subset worth detailed persona evaluation. The three-tier structure maintains a learning signal for all path qualities while focusing expensive compute on genuinely promising explanations.

4 Clustering User Feedback

Following the embedding process described in Phase I, we applied three complementary clustering algorithms to identify coherent interpretive profiles within each dataset. We visualized the clustering results using both PCA and t-SNE projections for dimensional reduction to verify cluster quality.

We systematically evaluated cluster solutions ranging from $k = 2$ to $k = 5$ using multiple internal validity metrics. The clustering revealed lower absolute silhouette scores (0.12 at $k = 2$), reflecting the greater heterogeneity in this combined task dataset. Despite the modest silhouette values, the convergence of all four validity metrics—silhouette score, inertia elbow, Davies-Bouldin minimum, and Calinski-Harabasz maximum—consistently pointed to a two-cluster solution.

Notably, the clustering assignments demonstrated remarkable stability across algorithms. The K-Means, Hierarchical Agglomerative, and HDBSCAN methods produced highly concordant partitions across all visualizations. This algorithmic agreement strengthens confidence in the natural groupings within our expert feedback data, as illustrated in Figure 3.

No responses were flagged as noise or isolated singletons, so all participants were directly assigned to one of the clusters, ensuring comprehensive coverage while maintaining cluster coherence.

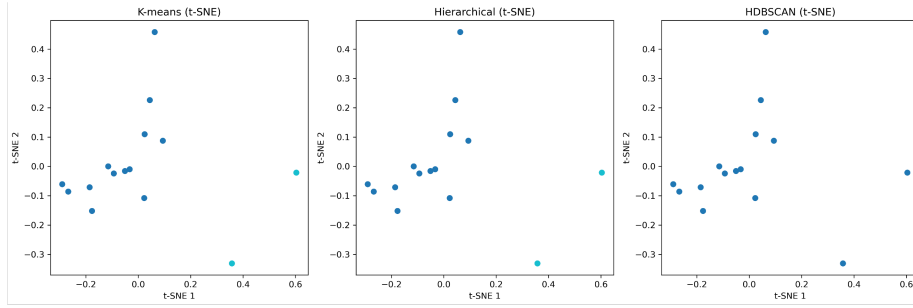


Figure 3: Visualization of clustering results. Each panel shows partitions produced by different algorithms (K-Means, Hierarchical, HDBSCAN) with embeddings projected via t-SNE.

5 Persona Assignment

How should explanations help you decide?

Choose the option that fits you better.

Q1

A detailed step-by-step explanation.

A streamlined explanation with fewer steps.

Q2

I do not need high-level concepts, I am happy with seeing only a detailed explanation.

I prefer to see high-level concepts that frame why things belong together.

Q3

Explanations should be based on specific relations such as "inhibits" or "binds".

Explanations can also include less specific relations such as "treats" or "causes".

Figure 4: Profiling questionnaire used to match participants to personas. For each question, the first response option (purple) aligns with Elena’s preferences, while the second option (green) aligns with Leo’s preferences. Response order was randomized for ordering effects.

Table 2: Mapping of profiling questions to persona characteristics based on expert feedback analysis

Question	Elena	Leo
Q1: Mech- anistic Detail	<i>“A detailed step-by-step explanation”</i> Supporting statements: - “combine therapeutic, genetic, and adverse-effect details for depth” - “combine multiple mechanistic or genetic details to reinforce direct drug-disease links”	<i>“A streamlined explanation with fewer steps”</i> Supporting statements: - “offer multiple paths for completeness while limiting their number to avoid excessive complexity” - “concise explanations that link drugs by shared treatment lines”
Q2: Ex- planation Complex- ity	<i>“I do not need high-level concepts, I am happy with seeing only a detailed explanation”</i> Supporting statements: - “explicitly state direct, clinically established drug-disease links” - “avoid generic class or side-effect links, deliver a clear therapeutic rationale”	<i>“I prefer to see high-level concepts that frame why things belong together”</i> Supporting statements: - “use ontological expansion to clarify connections between claims” - “leverage ontology for non-obvious or indirect relations”
Q3: Decision- making Context	<i>“Explanations should be based on specific relations such as ‘inhibits’ or ‘binds’”</i> Supporting statements: - “prefer ‘binds’ to ‘causes’ for drug-target interactions” - “eliminate vague predicates like ‘causes,’ ‘includes,’ or ‘participates’” - “avoid ambiguous ‘causes’ relations”	<i>“Explanations can also include less specific relations such as ‘treats’ or ‘causes’”</i> Supporting statements: - “relations...already covered by ‘causes’ or ‘treats’” (accepting them) - “ontological expansion specifically to indirect relations—those not already covered by ‘causes’ or ‘treats’”

6 User Study

We recruited 22 participants with backgrounds in biomedical and health-related sciences, including PhD students, postdocs, and researchers from both academic and industry settings (see Table 3). Participants reported their main area of expertise, their academic background across different fields, as well as the level of expertise in molecular biology, knowledge graphs, and AI systems.

The participant pool consisted primarily of early-career researchers, with 54.5% ($n = 12$) aged 25-29 years, followed by 18.2% ($n = 4$) aged 30-34, 13.6% ($n = 3$) aged 40-44, and smaller representations in the 20-24 (9.1%, $n = 2$) and

45-54 (4.5%, $n = 1$) age ranges. This age distribution reflects the typical composition of academic research environments where PhD students and postdoctoral researchers constitute the majority of the workforce. Analysis of primary expertise areas revealed three major groups: 58.3% from life sciences backgrounds (including biochemistry, biotechnology, microbiology, and medicine), 20.8% from computer science and AI, and 20.8% from explicitly interdisciplinary fields (bioinformatics, biomedical engineering, and health informatics).

Table 3: Participant demographics, background categories, main expertise areas, and domain knowledge ($n = 22$).

Category		Distribution
Age Range		20–24: 2; 25–29: 12; 30–34: 4; 40–44: 3; 45–54: 1
Persona Assigned		Leo: 12; Elena: 10
Primary Background	Back-	Life and Health Sciences: 15 (68.2%) CS/AI-heavy with Biomedical Experience: 3 (13.6%) Hybrid Computational-Biology: 4 (18.2%)
Main Expertise Area		Biochemistry: 2 Animal Cell Technology: 1 Microbiology: 2 Data Analytics: 1 Computer Science: 2 Biochemistry and Biotechnology: 1 Biotechnology: 2 Biochemistry, Computer Science: 1 Biomedical Engineering: 1 Bioinformatics, Computer Science: 1 Medicine: 1 Health and Bioinformatics: 1 Pharmacy: 1 Forensic Science: 1 Bioinformatics: 1 Veterinary Medicine: 1 Marine Biology: 1 Biological Engineering: 1
Domain Knowledge		
Knowledge Graphs		Expert: 5; Competent: 2; Novice: 11; No Knowledge: 4
AI Systems		Expert: 1; Competent: 5; Novice: 12; No Knowledge: 4
Molecular Biology		Expert: 5; Competent: 11; Novice: 6

Participants reported varying levels of formal training across computer science and life sciences. For computer science, 50% ($n = 11$) reported only *Some Knowledge*, while 36.4% ($n = 8$) held advanced training at MSc level or higher. Life sciences backgrounds showed stronger formal training overall, with 77.3% ($n = 17$) of participants holding advanced degrees or equivalent experience (MSc level or higher).

Participants also self-reported expertise that varied considerably across three critical domains. For knowledge graphs, 68.2% of participants rated themselves as *Novices* ($n = 11$) or having *No Knowledge* ($n = 4$). Similarly, AI systems expertise was predominantly at *Novice* level (54.5%, $n = 12$) or absent (18.2%, $n = 4$). In contrast, molecular biology expertise was substantially higher, with 72.7% rating themselves as *Competent* (50.0%, $n = 11$) or *Expert* (22.7%, $n = 5$) level, confirming a successful recruitment of domain experts. This distribution

aligns with our target user population: biomedical domain experts who interpret scientific evidence but are not necessarily specialists in the computational methods that generate it. In drug discovery practice, researchers can evaluate mechanistic explanations without requiring deep technical knowledge of knowledge graphs or AI systems. Our recruitment, therefore, intentionally prioritized biomedical expertise over graph-based technical proficiency, as we aim to evaluate how domain experts engage with scientific explanations.

The study was conducted using the online survey tool Tally¹. Each participant completed a 13-page form that included consent, a short demographic item (age group), and three profiling questions for persona assignment, followed by the main evaluation tasks (individual explanation ratings, comparative assessments, and text-versus-visual explanation preferences) and concluding with optional general open-ended feedback. We selected drug repurposing as the evaluation setting because it is an interpretively demanding task, requiring reasoning across heterogeneous biomedical entities. Each participant evaluated ten drug-repurposing hypotheses, viewing two explanations per hypothesis: one standard REx explanation and one adaptive explanation. We chose REx as the uniform baseline to ensure a controlled comparison under identical architecture and training conditions. To control for ordering effects, we randomized the presentation order of conditions across participants.

7 Wilcoxon Signed-Rank Tests

Overall, adaptive explanations significantly outperformed uniform explanations for relevance ($W = 3170.5, p < .001, r = 0.076$) and validity ($W = 2637.0, p < .001, r = 0.083$), though completeness showed no significant difference ($W = 4518.5, p = .107$). When analyzed separately by persona, Elena-aligned explanations showed significant improvements across all three metrics (relevance: $W = 295.5, p < .001$; completeness: $W = 728.0, p = .016$; validity: $W = 291.5, p < .001$), while Leo-aligned explanations improved significantly only for relevance ($W = 1459.5, p = .010$) and validity ($W = 1097.0, p = .024$), with no improvement in completeness ($W = 1592.5, p = .890$). While statistically significant, the effect sizes were small ($r < 0.10$), indicating modest but consistent improvements.

8 Posterior Credibility Analysis for Leo

Given Leo’s small formative base ($n=2$) yet substantial assignment in the user study ($n=12$), we conducted a posterior analysis to verify that Leo captures a coherent evaluative strategy rather than an artifact of limited data.

We prompted GPT-4o-mini with Leo’s persona narrative to evaluate the same 10 drug repurposing explanations assessed by the 12 Leo-assigned participants. For each hypothesis, the persona rated both the REx (non-adaptive)

¹<https://tally.so>

and Leo-conditioned explanations on relevance, completeness, and validity using identical scales and instructions as human participants. We then computed Pearson correlations between Leo persona ratings and aggregated ratings from the 12 Leo-assigned participants.

Table 4 presents the correlation analysis. Leo shows statistically significant correlations with participant ratings across all three dimensions, confirming that despite its minimal formative base, the persona captures a meaningful interpretive stance shared by the assigned participants.

Table 4: Posterior credibility analysis: Pearson correlations between Leo persona ratings and aggregated ratings from Leo-assigned participants (n=12).

Dimension	r	p-value
Validity	0.56	.010
Relevance	0.50	.024
Completeness	0.47	.038

9 LLM Prompt for Persona Synthesis

The following prompt was used with OpenAI’s o3-pro model to synthesize natural language personas from clustered user feedback. The model was instructed to follow this structure and generate both a narrative profile and supporting evidence.

Below are individual user feedback records from a group who share similar preferences. Each user has provided their background and detailed preferences.

Core Requirements:

- Analyze frequency patterns: Count how many users mention each theme
- Weight by user count: Prioritize themes mentioned by multiple users
- Flag weak signals: Mark any trait mentioned by <25% of users as potentially unreliable
- Create coherent character narratives: Write profiles as if describing a specific individual, not aggregated user research
- Validate with evidence: Support all persona traits with concrete user counts and percentages

Key Instruction:

Write the Profile section as if describing a real person’s preferences and behaviors.

Never mention "users", "this group", or research language within the Profile.

Save all analytical language for the Evidence Summary.

User Data Format:

User ID: [user_id]

Background: [user_background]
 Preferences: [full_preference_text]

Analysis Process:

Step 1: Frequency Count

- List each distinct preference theme
- Count how many users (and %) mention each theme
- Identify the top 3-4 most common themes (these become core traits)

Step 2: Evidence Quality Check

- Strong support: 40%+ of users
- Moderate support: 25-40%
- Weak signals: <25%

Step 3: Background Pattern Analysis

- Note background diversity without over-characterizing

Persona Structure:

Name: Descriptive name (avoid titles unless 60%+ of users explicitly hold those roles)

Tagline: One sentence capturing the strongest supported theme (40%+ users)

Profile:

- Lead paragraph: Core traits (40%+ support)
- Secondary paragraph: Additional values (25-40%)
- "Also values..." section: Traits in 15-25% range
- Background note: Brief mention of role diversity

Writing Guidelines:

Good: "Persona prefers direct pathways..."

Good: "She values concise summaries..."

Bad: "Users in this group prefer..."

Bad: "This persona represents users who..."

Evidence Summary:

- Core traits (40%+): List with counts and percentages
- Secondary traits (25-40%): List with counts
- Weak signals (<25%): List or omit
- Background composition: Summary
- Potential concerns: Biases, outliers, alignment issues

User Records:

{individual_user_records}

Output Format:

Name: [Descriptive name]

Tagline: [Strongest theme]

Profile: [Narrative description]

Evidence Summary: [Analytical details]

10 LLM Prompt for RL Reward

At the end of each RL rollout, the active persona, in this example Elena, rates each path according to the following LLM prompt:

```
You are evaluating explanation paths from the perspective of the
following persona:
"Name: Elena
Tagline: Trusts nothing but rock-solid mechanistic logic.
Profile: Elena weighs every therapeutic claim against hard
      mechanistic evidence. She expects an explanation to start with
      a clinically proven drug-disease link or a well documented mode
      of action-microtubule inhibition, T-cell suppression, a
      binding IC(50) in hand. Side-effect coincidences, vague "causes
      " predicates, or generic class labels make her skeptical unless
      an ontological note shows exactly why that class matters
      biologically. She welcomes extra layers-genetic markers,
      adverse-effect profiles, ontology tags-only when they tie
      directly to the core mechanism and keep the path lean.
      Superficial complexity or redundant paths lose her attention; a
      handful of well-justified edges beats a tangled web every time
      .
Also values... Occasional ontological expansions that rescue an
      otherwise hidden mechanistic clue, provided they do not swamp
      the narrative.
Background note: Elena's community is split between bench
      researchers and AI modelers, and she holds both to the same
      rigorous standard of scientific plausibility."xs

Score EACH path individually on three criteria:
1. Scientific Validity (V): 1-5. Scientific correctness,
   plausibility, and coherence based on biomedical knowledge.
2. Completeness (C): 1-5, where 3 is ideal. 1 = too simple, 5 = too
   complex. Reward paths that are sufficiently detailed without
   overload.
3. Relevance (R): 1-5. Usefulness for understanding why the
   prediction matters and how it connects to the task.

Instructions:
- Score each path individually on V, C, and R.
- Output strictly in JSON format as an array of objects:
  [{"id": <path id>, "validity": <int>, "completeness": <int>, "
    relevance": <int>}]
- Only use provided ids. Do not add or omit any.
```

11 LLM Prompts for Graph Verbalization

The following are the exact prompts used with the vision-language model to verbalize KG-based drug repurposing explanations. Three versions were implemented (brief, standard, comprehensive), with and without persona conditioning. Below we show the *standard* detail-level prompts.

11.1 Generic Verbalization Prompt

Write a natural, flowing description of the connections in this drug repurposing explanation.
 The graph shows paths from [DRUG] to [DISEASE].
 Cover the key relationships in 4-6 COMPLETE sentences that flow together naturally.
 If there are grey colored nodes, mention these as ontological expansions (lowest common ancestors).
 Write as connected prose, not as bullet points or lists.
 Use transitional phrases to connect ideas smoothly.

CRITICAL REQUIREMENTS:

- Do NOT describe the explanation in generic ways, make sure all sentences include a specific reference to the elements in the paths
- Every sentence MUST be grammatically complete with proper endings
- Do NOT write partial or incomplete sentences
- Do NOT include overall statements like "This demonstrates" or "Overall"
- Do NOT write concluding or summary statements
- Focus ONLY on describing the specific connections you see
- Continue describing pathways and relationships without meta-commentary
- Stop naturally after covering the key relationships

11.2 Persona-Conditioned Verbalization Prompt

Adopt this persona completely:
 [PERSONA_TEXT_INSERTED_HERE]

Write a natural, flowing description of the connections in this drug repurposing explanation.
 The graph shows paths from [DRUG] to [DISEASE].
 Cover the key relationships in 4-6 COMPLETE sentences that flow together naturally.
 If there are grey colored nodes, mention these as ontological expansions (lowest common ancestors).
 Write as connected prose, not as bullet points or lists.
 Use transitional phrases to connect ideas smoothly.

CRITICAL REQUIREMENTS:

- Do NOT describe the explanation in generic ways, make sure all sentences include a specific reference to the elements in the paths
- Every sentence MUST be grammatically complete with proper endings
- Do NOT write partial or incomplete sentences
- Do NOT include overall statements like "This demonstrates" or "Overall"
- Do NOT write concluding or summary statements
- Focus ONLY on describing the specific connections you see
- Continue describing pathways and relationships without meta-commentary
- Stop naturally after covering the key relationships

You must embody this persona's specific priorities, standards, and interests.

Write ONLY about what this persona would find valuable and relevant

Filter the entire description through this persona's lens - include details they would care about and completely omit information they would consider irrelevant or superficial.

Match this persona's preferred depth, focus, and style.

Be selective and discriminating based on what matters to THIS specific viewpoint.

Never mention any names from the perspective description.

REMEMBER: Complete all sentences properly.

Do NOT describe the explanation in generic ways, make sure all sentences include a specific reference to the elements in the paths.